

ABSTRACT

Surveillance video quality degrades severely under adverse weather conditions such as fog, rain, and snow, and single image restoration models struggle to generalize across weather types.

This paper presents RoadClear, a system that employs MobileNetV3-Small as a weather-aware front-end module for real-time weather classification. An inter-frame skipping scheduler dynamically loads the appropriate restoration model—dehazing, deraining, or desnowing—based on the classification result, and ONNX Runtime is adopted to accelerate YOLO detectors and all restoration models. Experiments on synthetically degraded DETRAC videos demonstrate that enhancement improves small-target Recall by 31.19% under fog, mAP@0.95 by 128.18% under rain, and high-confidence detections by 9.01% under snow. Ablation studies confirm that applying a single restoration network across weather types degrades mAP@0.95 by up to 75.0%, validating the necessity of weather-specific models. A "pixel-semantic discrepancy" phenomenon observed under fog reveals a systematic misalignment between pixel-level metrics and task-level performance.

Based on these methods, a B/S-architecture video enhancement visualization system is developed using Vue and FastAPI, integrating weather-aware scheduling, multi-model restoration, parameter control, and detection evaluation, demonstrating the feasibility of the proposed approach in practical engineering applications.

Keywords: traffic surveillance; video enhancement; image dehazing; image deraining; image desnowing; weather awareness; model scheduling; deep learning

CONTENT

摘 要	II
ABSTRACT	III
CHAPTER 1: INTRODUCTION.....	1
1.1 Background	1
1.2 The Purpose and Significance of Project	2
1.3 About system.....	3
1.4 The Organization of This Article	3
1.5 Chapter Summary.....	3
CHAPTER 2: RELATED WORK	4
2.1 Weather Classification	4
2.1.1 Traditional Feature-based Methods	4
2.1.2 Deep Learning-based Methods	4
2.2 Single Image Restoration	5
2.2.1 Image Dehazing	5
2.2.2 Image Deraining	5
2.2.3 Image Desnowing	6
2.3 Video Enhancement for Surveillance.....	6
2.3.1 Temporal Consistency	6
2.3.2 Real-time Processing	6
2.3.3 Multi-weather Model Coordination.....	6
2.3.4 Task-driven Evaluation.....	7
2.4 Chapter Summary.....	7
CHAPTER 3: PROPOSED METHOD FOR VIDEO ENHANCEMENT AND SCHEDULING.....	8
3.1 Overall Idea	8
3.2 Intelligent Weather Classification and Scheduling Module.....	9
3.2.1 Design of Weather Classification Network	9
3.2.2 Scheduling Strategy	10
3.3 Weather Degradation Restoration Networks.....	11

3.3.1 Dehazing Network	12
3.3.2 Deraining Network	12
3.3.3 Desnowing Network	12
3.4 Video Stream Optimization and Inference Acceleration	12
3.4.1 Temporal Consistency Processing	13
3.4.2 ONNX Inference Acceleration	13
3.5 Downstream Task Evaluation Module	14
3.5.1 Vehicle Detection	14
3.5.2 License Plate Detection	15
CHAPTER 4: Experiments and Analysis.....	16
4.1 Datasets and Experimental Setup	16
4.2 Evaluation Metrics	17
4.3 Performance of Weather Classification Module	20
4.4 Performance of Individual Restoration Networks.....	21
4.4.1 Rainy Weather Enhancement Performance	22
4.4.2 Foggy Weather Enhancement Performance	23
4.4.3 Snowy Weather Enhancement Performance	25
4.5 Ablation Study.....	26
4.6 Chapter Summary.....	28
CHAPTER 5: SYSTEM DESIGN AND DEVELOPMENT	29
5.1 Requirement Analysis	29
5.1.1 Functional requirements	29
5.1.2 Non-functional requirements.....	29
5.2 System Context	30
5.2.1 Overall Architecture	30
5.2.2 Module Interaction Flow	31
5.3 System Interface.....	32
5.3.1 Main Interface.....	32
5.3.2 Video Enhancement Interface	33
5.3.3 Algorithm Showcase Interface	34

5.3.4 Parameter Console Interface.....	35
5.3.5 Evaluation Interface.....	37
5.4 Chapter Summary.....	38
CHAPTER 6: CONCLUSION AND FEATURE WORK.....	39
6.1 Conclusion.....	39
6.2 Feature Work.....	40
Reference	41
Acknowledgment.....	45

CHAPTER 1: INTRODUCTION

1.1 Background

As urbanization accelerates and vehicle ownership rises, traffic congestion and accidents place growing demands on Intelligent Transportation Systems (ITS). Traffic surveillance networks generate massive video data daily, from which computer vision algorithms extract essential information such as traffic flow and vehicle trajectories [1]. Their effectiveness, however, depends heavily on video quality—severely compromised by adverse weather. Haze reduces contrast and visibility, rain introduces streaks and motion blur, and snow causes local occlusion and additive noise [2], threatening downstream tasks such as license plate recognition and vehicle identification that rely on fine-grained visual details.

This challenge has driven extensive research in image restoration. While traditional methods such as histogram equalization and prior-based dehazing offer limited effectiveness [3], deep learning—particularly CNNs and Transformers—has significantly advanced the field. Architectures including AOD-Net, PReNet, and TransWeather have achieved state-of-the-art results in dehazing, deraining, and desnowing [4][5][6].

Despite this progress, three limitations persist. First, existing methods target single weather types and lack adaptive scheduling; real-world streams may encounter shifting conditions that a single network cannot handle. Second, evaluation relies on pixel-level metrics such as PSNR and SSIM, neglecting downstream task performance. Third, weather recognition, restoration, and stream optimization are rarely integrated into an end-to-end system that satisfies real-time, stability, and deployability requirements.

To overcome the above limitations, this paper presents RoadClear, a traffic surveillance video enhancement system for adverse weather. A lightweight classification network performs real-time weather recognition, an intelligent scheduler routes frames to weather-specific restoration networks, and ONNX acceleration with temporal consistency optimization ensures real-time, stable processing. Enhanced frames are evaluated through downstream YOLO detection and license plate recognition, validating the practical benefits of multi-weather restoration for ITS perception.

1.2 The Purpose and Significance of Project

The significance of this project lies in its systems-engineering perspective: it integrates deep learning-based image restoration techniques with practical traffic surveillance requirements to construct a complete pipeline spanning weather recognition, intelligent scheduling, multi-weather restoration, video stream optimization, and downstream task validation. This not only enhances the reliability of traffic surveillance under adverse weather but also provides a referenceable engineering paradigm for the design and deployment of similar intelligent vision systems.

The project aims to design and implement an intelligent video enhancement system for adverse weather conditions, with the following specific objectives:

(1) Intelligent weather classification and scheduling. Train a lightweight convolutional neural network to recognize fog, rain, snow, and normal weather in real time, and design a scheduling strategy that adaptively selects the most appropriate restoration network based on the classified weather type.

(2) Multi-weather degradation restoration integration. Integrate three deep learning-based restoration networks—dehazing, deraining, and desnowing—into a unified framework, enabling effective quality enhancement under corresponding weather conditions.

(3) Real-time video stream optimization. Apply ONNX inference acceleration and temporal consistency processing to ensure sufficient real-time performance and inter-frame stability for practical deployment.

(4) Downstream task validation. Train YOLO-based object detection and license plate recognition models on DETRAC and CCPD datasets, respectively, and quantitatively compare detection outputs before and after enhancement to verify practical benefits for ITS perception tasks.

(5) System integration and engineering implementation. Develop a complete video enhancement system prototype using FastAPI and a modular frontend–backend architecture, providing a visual processing interface and result display to validate the engineering feasibility of the proposed approach.

1.3 About system

Back-end Language: Python

Back-end Framework: FASTAPI

Front-end Framework: Vue.js

Deep learning Framework: Pytorch

IDE: Visual Studio Code

GPU: GeForce RTX 4070 Ti SUPER

1.4 The Organization of This Article

Chapter 1: This chapter introduces the background, reviews related work, and outlines the main contributions and structure of this work.

Chapter 2: This chapter surveys existing methods in weather classification, image restoration, and downstream detection, identifying the limitations that motivate the RoadClear system.

Chapter 3: This core chapter presents the weather classification network, intelligent scheduling strategy, dedicated restoration networks, and downstream evaluation module, with extensive experiments and ablation studies.

Chapter 4: This chapter describes the system architecture, module interaction flows, and the frontend interface built with Vue and FastAPI.

Chapter 5: This chapter summarizes the main contributions, discusses current limitations, and outlines future research directions.

1.5 Chapter Summary

This chapter introduced the background of traffic surveillance under adverse weather, highlighting the degradation challenges posed by fog, rain, and snow, and outlined the project's five research objectives from a systems-engineering perspective, establishing the scope for the remainder of this work.

CHAPTER 2: RELATED WORK

2.1 Weather Classification

Weather classification aims to automatically identify the weather type (e.g., fog, rain, snow, clear) from an input image or video frame. It serves as the decision-making front-end in multi-weather enhancement systems.

This chapter reviews prior research in three areas that are essential to the design of RoadClear. The first area is weather classification, which serves as the decision-making front-end that determines which restoration model to invoke. The second area is single image restoration, encompassing dehazing, deraining, and desnowing, which together form the core enhancement capability of the system. The third area concerns the additional requirements that arise when applying these techniques to video, including temporal stability, real-time throughput, coordination among multiple specialized models, and task-oriented evaluation.

2.1.1 Traditional Feature-based Methods

Early weather classification methods relied on hand-crafted features. Color features analyze the statistical distribution of color channels [7], texture features capture structural differences [8], and gradient features measure edge information [9]. While these hand-crafted representations offer interpretability and low computational cost, their reliance on manually designed rules limits their ability to generalize across the diverse and complex visual patterns encountered in real-world weather conditions.

2.1.2 Deep Learning-based Methods

Convolutional neural networks (CNNs) have become the dominant approach for weather classification by automatically learning hierarchical features from raw pixels. WeatherNet introduced a CNN-based weather classification module that enables automatic selection of appropriate restoration networks [10]. In a recent comparative study, CNN, VGG, and Vision Transformer (ViT) models all achieved classification accuracies exceeding 90% on a ten-category weather dataset [11].

For real-time applications, lightweight architectures are essential. MobileNet [12] and EfficientNet [13] minimize computational overhead while maintaining competitive accuracy, making them suitable for video-rate processing. Vision Transformers [14]

and Swin Transformers [15] capture long-range dependencies for improved classification, though their computational cost remains a challenge for real-time deployment.

Despite this progress, most existing weather classification studies treat classification as an isolated task rather than integrating it into a complete enhancement pipeline with adaptive scheduling—a gap that RoadClear addresses.

2.2 Single Image Restoration

Single image restoration under adverse weather has been extensively studied, with specialized techniques developed for dehazing, deraining, and desnowing.

2.2.1 Image Dehazing

Prior-based dehazing methods leverage physical atmospheric scattering models and statistical priors about haze-free images. The Dark Channel Prior (DCP) [16] and the Color Attenuation Prior (CAP) [17] provide interpretable solutions but often fail when assumptions do not hold.

Deep learning has become the mainstream approach. DehazeNet [18] and MSCNN [19] were among the early CNN-based dehazing networks. AOD-Net proposed a lightweight architecture that directly estimates the clean image [4]. GridDehazeNet [20] explored attention-based multi-scale architectures, and AECR-Net [21] introduced contrastive learning for compact dehazing. Recent surveys provide comprehensive taxonomies of deep dehazing methods [22][23].

2.2.2 Image Deraining

Model-based deraining methods typically formulate rain removal as a sparse coding problem [24]. Deep learning has significantly advanced deraining. DerainNet pioneered CNN-based rain removal [25]. RESCAN introduced recurrent squeeze-and-excitation context aggregation [26]. PReNet proposed a progressive recurrent architecture [5]. A comprehensive survey by Su et al. categorizes deep deraining methods [27]. Recent works have explored attention mechanisms [28] and self-supervised approaches [29].

2.2.3 Image Desnowing

Snow degradation presents unique challenges due to particle accumulation, motion blur, and scene occlusion [30]. Traditional methods, such as the shooting algorithm for joint rain and snow removal [31], often fail to generalize.

DesnowNet introduced a context-aware network for snow removal [30]. HDCWNet combined wavelet domain processing with spatial domain convolutions [32]. More recently, context-aware coarse-to-fine architectures with contrastive learning [33] and Transformer-based methods [6] have achieved good results.

2.3 Video Enhancement for Surveillance

Applying enhancement to surveillance video introduces additional requirements beyond single image restoration: temporal consistency, real-time performance, multi-weather coordination, and task-driven evaluation.

2.3.1 Temporal Consistency

Per-frame enhancement without temporal modeling often produces flickering artifacts. Video enhancement methods have explored optical flow alignment [34], recurrent networks [35], and 3D convolution approaches [36]. More recently, pre-trained diffusion models have been adapted for temporally consistent video restoration [37]. Collaborative learning between adjacent frames through adversarial training and temporal loss functions has demonstrated improved visual quality [38].

2.3.2 Real-time Processing

Real-time video enhancement requires 25–30 FPS. Model compression techniques—including network pruning [39], quantization [40], and knowledge distillation [41]—reduce computational requirements. Lightweight architectures such as MobileNetV3 [42] and EfficientNet [13] provide efficient design patterns. Deployment frameworks including ONNX Runtime and TensorRT further enable efficient inference. A recent comparative analysis of inference frameworks on edge hardware confirmed significant impacts of framework selection [43].

2.3.3 Multi-weather Model Coordination

Practical systems must handle multiple weather conditions. Weather classification enables automatic selection of restoration algorithms. Sequential model application

addresses compound weather effects, and unified architectures like TransWeather [6] handle multiple weather types in a single model. Hybrid approaches combining classification with cascaded processing remain underexplored.

2.3.4 Task-driven Evaluation

Traditional metrics like PSNR and SSIM do not fully capture practical value. Task-driven evaluation frameworks measure improvements in downstream computer vision tasks [44]. Object detection metrics—precision, recall, and mAP—evaluate detection accuracy improvement. Rain removal in traffic surveillance has been studied to assess its impact on subsequent tasks [44]. TIENet proposed a task-oriented image enhancement network achieving AP improvements on foggy datasets [45]. End-to-end systems that jointly optimize enhancement and detection, such as TIENet [45], may achieve better task performance, though generalization remains a challenge.

2.4 Chapter Summary

This chapter reviewed three areas of related work. Weather classification methods have advanced from hand-crafted features to lightweight deep learning models, but are rarely integrated with downstream scheduling and restoration. Single image restoration has achieved impressive results for individual weather types through specialized deep architectures, yet these methods are developed in isolation. Video enhancement research has addressed temporal consistency, real-time processing, multi-weather coordination, and task-driven evaluation, but few systems combine all these components into a unified, deployable pipeline. These gaps motivate the RoadClear system presented in Chapter 3.

CHAPTER 3: PROPOSED METHOD FOR VIDEO ENHANCEMENT AND SCHEDULING

3.1 Overall Idea

This chapter proposes RoadClear, an intelligent enhancement system for road surveillance videos under adverse weather conditions. The system achieves adaptive enhancement of video quality under fog, rain, and snow conditions through the collaborative operation of weather classification, model scheduling, deep learning enhancement networks, and inference optimization modules. The core idea is: first, a lightweight convolutional neural network is used to identify the weather type of input video frames; then, corresponding weather degradation restoration networks (dehazing, deraining, or desnowing) are intelligently scheduled based on the recognition results; finally, real-time performance and smoothness of video enhancement are ensured through ONNX inference acceleration and temporal consistency processing.

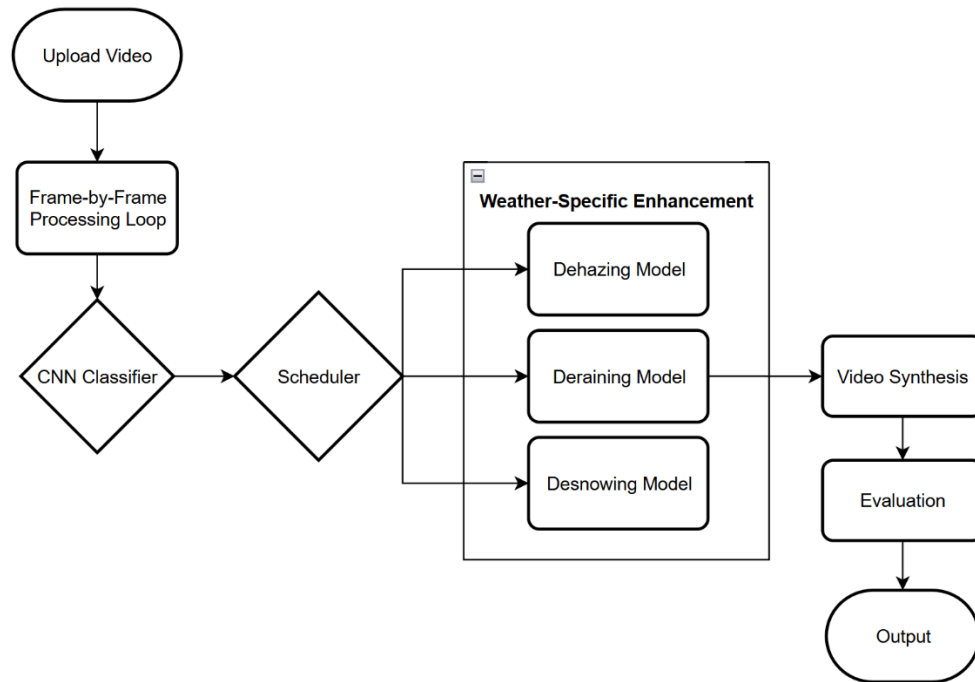


Figure 3-1. Traffic surveillance video enhancement system flowchart

3.2 Intelligent Weather Classification and Scheduling Module

The intelligent weather classification and scheduling module is the core decision-making component of the RoadClear system, responsible for identifying weather types in video frames and dynamically scheduling corresponding enhancement models. The design goal of this module is to minimize computational overhead while ensuring classification accuracy, achieving real-time video stream processing.

3.2.1 Design of Weather Classification Network

The proposed system adopts MobileNetV3-Small as the backbone architecture for the weather classification network. MobileNetV3 is a lightweight convolutional neural network specifically designed for mobile and embedded platforms. By combining depthwise separable convolutions with squeeze-and-excitation (SE) attention mechanisms, it achieves competitive classification accuracy while maintaining low computational complexity.

The weather classification network takes an RGB image of size $224 \times 224 \times 3$ as input and outputs a 3-dimensional probability vector corresponding to three weather categories: fog, rain, and snow. The architecture consists of the following components:

Feature extraction backbone. We employ the pre-trained MobileNetV3-Small as the feature extractor. It comprises a stack of inverted residual bottleneck blocks, each consisting of a 1×1 expansion convolution, a 3×3 depthwise convolution, and a 1×1 projection convolution. This design effectively decouples spatial and channel-wise feature learning, substantially reducing the parameter count and computational cost.

Attention mechanism. SE modules are integrated at key layers of the backbone. Each SE module performs global average pooling followed by a two-layer fully connected network to model inter-channel dependencies and adaptively recalibrate channel-wise feature responses, enhancing the network's sensitivity to informative features relevant to weather classification.

Classification head. The final fully connected layer of the backbone is replaced with a 3-way classification layer, and a Softmax activation function is applied to produce the predicted probability distribution over the three weather types.

The forward computation can be expressed as:

$$y = \text{Softmax}(W_c \times f_{\text{MobileNetV3}}(x) + b_c) \quad (1)$$

where $x \in \mathbb{R}^{224 \times 224 \times 3}$ denotes the input image, $f_{\text{MobileNetV3}}()$ represents the feature extraction function of the MobileNetV3 backbone, W_c and b_c are the weight matrix and bias vector of the classification layer, and $y \in \mathbb{R}^3$ is the output probability vector.

The model is trained using the cross-entropy loss function, defined as:

$$\mathcal{L} = - \sum_{i=1}^N \sum_{c=1}^3 y_{i,c} \log(\hat{y}_{i,c}) \quad (2)$$

where N denotes the batch size, $y_{i,c}$ is the one-hot encoded ground-truth label, and $\hat{y}_{i,c}$ is the predicted probability for class c .

The network is optimized using the Adam optimizer with an initial learning rate of 0.001 and a cosine annealing learning rate schedule. Training is conducted on the RSCM (Road Surveillance under Challenging weather and lighting conditions for autonomous driving) dataset, which contains road surveillance images captured under fog, rain, and snow conditions.

Figure 3-2 illustrates the overall architecture of the weather classification network and the feature extraction pipeline.

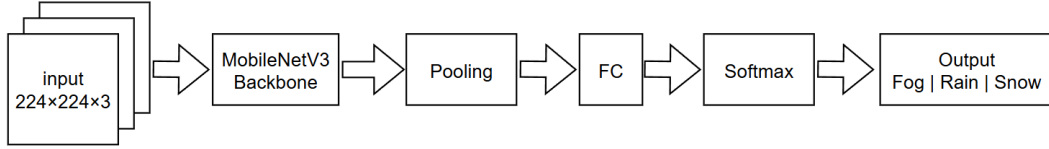


Figure 3-2. weather classification network architecture diagram

3.2.2 Scheduling Strategy

To balance classification accuracy with computational efficiency, we design an intelligent scheduling strategy based on inter-frame skipping. The core premise of this strategy is that weather conditions exhibit high temporal coherence across consecutive video frames. It is therefore unnecessary to perform weather classification on every individual frame; instead, a periodic sampling scheme can substantially reduce the invocation frequency of the classification network while preserving weather recognition accuracy.

The scheduler maintains per-stream state information, including a frame counter, the most recent classification result, and the active model chain. The algorithmic workflow is illustrated in the flowchart below.

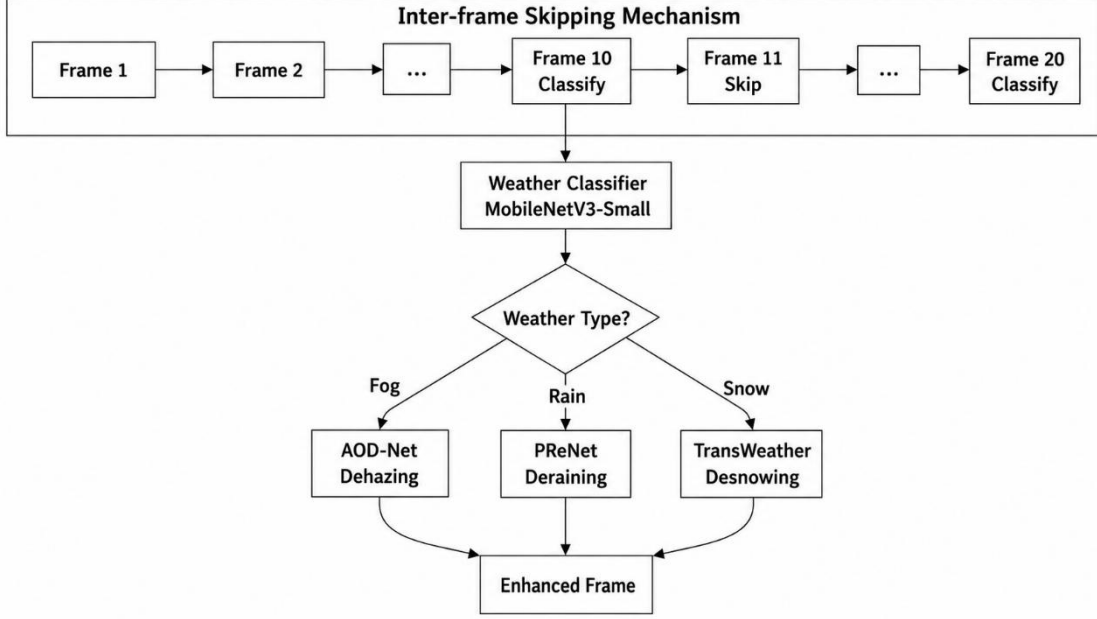


Figure 3-3. Workflow of the scheduling strategy with inter-frame skipping.

Based on the classification output, the scheduler selects the corresponding enhancement model according to the following mapping rules:

- 1) Fog $\rightarrow$ AOD-Net (dehazing network)
- 2) Rain $\rightarrow$ PReNet (deraining network)
- 3) Snow $\rightarrow$ TransWeather (desnowing network)

For composite weather scenarios, the system supports adaptive scheduling via a model cascading strategy. Users may also manually specify a model chain, such as aodnet $\rightarrow$ prenet, which applies dehazing followed by deraining. The cascading operation can be expressed as:

$$I_{enhanced} = M_2 \left(M_1(I_{input}) \right) \quad (3)$$

where I_{input} denotes the input frame, M_1 and M_2 are the first- and second-stage enhancement models, respectively, and $I_{enhanced}$ is the final enhanced output.

3.3 Weather Degradation Restoration Networks

The weather degradation restoration networks serve as the core processing modules of the RoadClear system, responsible for restoring images degraded under adverse weather conditions. Since the primary contribution of this work lies in system integration rather than the design of individual restoration models, we adopt three existing networks, each pre-trained for a specific weather type: AOD-Net for fog, PReNet for rain, and TransWeather for snow. These models are integrated as off-the-

shelf components, and their architectures are only briefly summarized below. Detailed descriptions can be found in the original publications.

3.3.1 Dehazing Network

AOD-Net (All-in-One Dehazing Network) is a lightweight dehazing model that reformulates the atmospheric scattering model into a single end-to-end trainable network. It directly learns the mapping from hazy images to clean images without explicitly estimating the transmission map or atmospheric light. We use the pre-trained weights provided by the original authors, trained on the ITS (Indoor Training Set) dataset. In our system, AOD-Net is triggered when the weather classification module detects foggy conditions.

3.3.2 Deraining Network

PReNet (Progressive Recurrent Network) is a recurrent deraining network that removes rain streaks progressively across multiple stages. Its recurrent architecture enables parameter sharing across stages, making it both parameter-efficient and effective at handling rain streaks of varying densities. We adopt the model pre-trained on the Rain100H dataset, which contains heavy rain scenarios. PReNet is activated when the weather classification module identifies rainy conditions.

3.3.3 Desnowing Network

TransWeather is a unified transformer-based architecture capable of handling multiple adverse weather conditions, including snow. Unlike AOD-Net and PReNet, which are weather-specific, TransWeather employs a shared encoder–decoder structure with learnable weather-type queries, allowing a single model to address multiple degradation types. For the snow restoration task, we use the model trained on the Snow100K dataset. In our system, TransWeather is invoked when snowy conditions are detected, and it also serves as the fallback option for composite weather scenarios due to its multi-weather capability.

3.4 Video Stream Optimization and Inference Acceleration

To meet the real-time processing requirements of road surveillance videos, this system optimizes video stream processing from two aspects: temporal consistency and inference acceleration, while ensuring enhancement quality.

3.4.1 Temporal Consistency Processing

A key challenge in video enhancement is temporal inconsistency. Since deep learning models process each frame independently, subtle variations may arise in the enhancement results of consecutive frames, leading to perceptible flickering artifacts that degrade the viewing experience.

Temporal inconsistency stems primarily from three sources: (1) stochasticity inherent in model outputs—even visually similar input frames may elicit slightly different network responses; (2) weather classification instability—when conditions lie near a decision boundary, the classifier may oscillate between categories, triggering different enhancement models in rapid succession; and (3) variations in illumination and object motion, which introduce genuine inter-frame differences that can be amplified by independent per-frame processing.

To mitigate these issues, our system adopts the following strategies:

Inter-frame skipping. As described in Section 3.2.2, the scheduling strategy classifies weather only at designated key frames and reuses the most recent classification result for intermediate frames. This ensures that the same enhancement model is applied throughout a short temporal window, preventing rapid model switching.

State retention. The scheduler maintains per-stream state information, including the last detected weather type, the active model chain, and a frame counter. This mechanism guarantees that classification decisions persist across consecutive unclassified frames, enforcing short-term temporal stability.

Together, these two mechanisms effectively suppress flickering caused by model switching and classification jitter. It should be noted, however, that they do not directly address pixel-level inconsistency between independently enhanced frames—an inherent limitation of single-frame enhancement methods.

3.4.2 ONNX Inference Acceleration

To accelerate model inference, the system adopts ONNX Runtime as the inference engine. ONNX (Open Neural Network Exchange) is an open format that enables models trained with frameworks such as PyTorch to be exported to a unified intermediate representation. ONNX Runtime, developed by Microsoft, applies a suite of graph-level optimizations—including operator fusion, constant folding, and memory reuse—to deliver significant inference speedups across platforms.

Three optimization levers are configured in our system: (1) Graph optimization level is set to the maximum (ORT_ENABLE_ALL), enabling all default optimization passes. (2) Execution provider is selected automatically based on available hardware: CPUExecutionProvider for CPUs and CUDAEExecutionProvider for GPUs. (3) Thread configuration adjusts the number of intra-op and inter-op threads to fully utilize multi-core processors. Additionally, optional INT8 quantization is available to further reduce model size and memory footprint, though it was not enabled in the experiments reported here.

3.5 Downstream Task Evaluation Module

3.5.1 Vehicle Detection

Vehicle detection is a fundamental task in traffic surveillance and serves as the primary downstream evaluation benchmark in this study. We train a four-class vehicle detector on the UA-DETRAC dataset to assess how weather enhancement affects detection performance. The detector is based on the YOLOv8 architecture, which offers a favorable balance between accuracy and inference speed and supports ONNX export for seamless system integration. The trained model distinguishes four vehicle categories: car, bus, van, and others. Dataset details are provided in Section 3.6.1.

The training configuration is summarized in Table 3-1.

Table 3-1 Training configuration of the DETRAC vehicle detector.

Parameter	Setup
Epochs	50
Optimizer	Adam
Initial learning rate	0.01
Learning rate schedule	Cosine annealing
mAP@0.5	0.785
mAP@0.5–0.95	0.645
Precision	0.814
Recall	0.690

Training converged steadily, with the most rapid loss reduction within the first 10 epochs and stabilization after approximately 30 epochs. These results indicate satisfactory detection performance and establish a reliable baseline for downstream evaluation.

3.5.2 License Plate Detection

License plate recognition is another desirable capability in traffic surveillance. As a supplementary evaluation, we train a YOLOv8-based license plate detector on the CCPD (Chinese City Parking Dataset), which contains over 250,000 images covering diverse scenes and weather conditions, including normal, blurred, and challenging weather scenarios. This allows us to examine whether weather enhancement can recover plate-level details for recognition.

The training configuration is summarized in Table 3-2.

Table 3-2. Training configuration of the CCPD license plate detector.

Parameter	Setup
Epochs	50
Optimizer	Adam
Initial learning rate	0.01
Learning rate schedule	Cosine annealing
mAP@0.5	0.791
mAP@0.5–0.95	0.791
Precision	~1.0
Recall	~1.0

The model reached an mAP@0.5 of approximately 0.78 by epoch 10 and stabilized by epoch 50. The exceptionally high metrics reflect the simplicity of the CCPD task and confirm that the detector is not a bottleneck in the evaluation pipeline.

However, we note a practical limitation: the surveillance cameras operate at an overhead angle with low resolution, so license plates occupy very few pixels. This module therefore serves as a proof-of-concept. Integrating camera zoom or region-of-interest cropping is left as future work.

CHAPTER 4: Experiments and Analysis

This section validates the effectiveness of the RoadClear system through experiments, conducting comprehensive evaluations from multiple dimensions including weather classification accuracy, performance of individual restoration networks, ablation studies, and comparative experiments.

4.1 Datasets and Experimental Setup

The datasets used in this study are summarized in Table 3-3. They serve three purposes: (1) weather classification—RSCM for training and DAWN for testing; (2) downstream vehicle detection—DETRAC was used both to train the YOLOv8 detector and, after synthetic degradation with rain, fog, and snow, to construct the test set; and (3) license plate detection—CCPD (250,000+ images) for training the YOLOv8 plate detector. Restoration network pre-training details (Rain100H, ITS, Snow100K) are omitted here and referenced in Section 3.3.

Table 4-1. Summary of datasets used in experiments.

Purpose	Dataset	Role	Sample Size	Description
Weather Classification	RSCM	Training	10,000 per type	Real-world road surveillance
	DAWN	Testing	~700	Independent real weather dataset
Pre-training (Restoration)	Rain100H	Pre-training	—	PRNet weights
	Snow100K	Training	~100,000	TransWeather weights
	ITS	Pre-training	—	AOD-Net weights
Downstream Evaluation	CCPD	Training (YOLO)	250,000+	License plate detection
	DETRAC (clean)	Training (YOLO)	~68,000	Four-class vehicle detection
	DETRAC+Synthetic	Testing	2,500 frames	20 scenes $\times$ 5 s $\times$ 25 fps

All experiments were conducted on a desktop workstation running Windows 11, with the configuration detailed in Table 4-1. The chosen GPU provides 16 GB of video memory, which is sufficient for the batch processing of 224×224 resolution frames used throughout this study. All deep learning models were implemented in PyTorch and accelerated via ONNX Runtime. No distributed training or cloud computing resources were required, ensuring that the entire pipeline remains deployable on a single consumer-grade machine.

Table 4-2. Experimental environment configuration.

Category	Item	Specification
Hardware	CPU	Intel i5-13600KF (8 cores, 16 threads)
	GPU	NVIDIA RTX 4070 Ti SUPER (16 GB VRAM)
	RAM	64 GB DDR4
Software	Operating System	Windows 11
	Python	3.10
	CUDA	12.4
	PyTorch	2.4.1
Key Libraries	ONNX Runtime	1.17
	OpenCV	4.13
	YOLO (Ultralytics)	8.0

4.2 Evaluation Metrics

This system adopts a multi-dimensional evaluation metric system, including image quality metrics, downstream task metrics, and performance metrics.

(1) Image Quality Metrics

To evaluate the pixel-level fidelity of enhanced images, we employ the following four metrics: PSNR, SSIM, Contrast and Noise Suppression Rate.

PSNR quantifies the pixel-wise error between the enhanced image and the clean reference image, serving as the most widely used objective quality metric in image restoration. It is defined as:

$$PSNR = 10 \times \log_{10} \left(\frac{MAX_I^2}{MSE} \right) \quad (4)$$

Where MAX_I^2 is the maximum possible pixel value (255 for 8-bit images) and MSE is the Mean Squared Error between the restored and ground-truth images. Higher

PSNR values indicate better restoration quality, with typical values for restored images ranging from 15 dB to 35 dB.

SSIM measures perceptual similarity between two images across three dimensions—luminance, contrast, and structure—offering better alignment with human visual perception than PSNR. SSIM ranges from 0 to 1, with values closer to 1 indicating higher structural similarity. It is defined as:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (5)$$

Contrast quantifies the degradation of global dynamic range caused by adverse weather, expressed as the standard deviation of pixel values:

$$Contrast = \sqrt{\frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W (I_{i,j} - \mu)^2} \quad (6)$$

μ denotes the mean pixel value. The contrast degradation ratio reflects the severity of weather-induced degradation, while the contrast recovery ratio measures the restoration capability of the enhancement system.

The Noise Suppression Rate evaluates the effectiveness of enhancement in suppressing high-frequency noise, defined as:

$$Noise\ Suppression = \left(1 - \frac{\sigma_{enhanced}^2}{\sigma_{degraded}^2}\right) \times 100\% \quad (7)$$

$\sigma_{enhanced}^2$ and $\sigma_{degraded}^2$ denote the noise variance of the enhanced and degraded images. Positive values indicate effective noise suppression, while negative values suggest that the enhancement process introduces additional high-frequency artifacts.

(2) Downstream Task Metrics

The ultimate objective of this system is to improve vehicle detection and license plate recognition accuracy in traffic surveillance videos under adverse weather conditions. Accordingly, we adopt the following four downstream task metrics as the core evaluation criteria.

Mean Confidence reflects the average certainty of the detector across all detection results, defined as:

$$Mean\ Confidence = \frac{1}{N} \sum_{i=1}^N c_i \quad (8)$$

N is the total number of detection boxes and c_i is the confidence score of the i -th box. Higher values indicate that the detector is more certain about the targets in the enhanced images.

High-Confidence Detection Count counts the number of detection boxes with confidence scores no less than 0.7. Compared with the total detection count, it filters out low-confidence false positives and better reflects system usability in real deployment scenarios.

DCCR measures the relative change in the number of detections before and after enhancement:

$$DCCR = \frac{N_{enhanced} - N_{degraded}}{N_{degraded}} \times 100\% \quad (9)$$

$N_{degraded}$ and $N_{enhanced}$ denote the total number of detection boxes before and after enhancement, respectively. Positive values indicate an increase in detections after enhancement, while negative values indicate a decrease.

The Recovery Rate is a comprehensive downstream evaluation metric proposed in this work, quantifying the proportion of weather-induced performance loss that is successfully restored by the enhancement system:

$$RR = \frac{M_{enhanced} - M_{degraded}}{M_{clean} - M_{degraded}} \times 100\% \quad (10)$$

M_{clean} , $M_{degraded}$ and $M_{enhanced}$ denote the downstream metric values on clean, weather-degraded, and enhanced videos. $RR = 100\%$ indicates full recovery to the clean baseline, $RR > 0\%$ indicates positive enhancement effects, and $RR < 0\%$ indicates that enhancement introduces further degradation.

Recall by Object Size. To conduct a fine-grained evaluation of the enhancement system's recovery capability for objects of different scales, we stratify detected objects into three tiers based on bounding box pixel area: small (area $< 32 \times 32$ pixels), medium ($32 \times 32 \sim 96 \times 96$ pixels), and large ($> 96 \times 96$ pixels). Recall is computed independently for each tier.

(3) Performance Metrics

To assess the practical deployability of the system, we record the following three performance metrics:

- FPS (Frames Per Second): The number of video frames processed per second, measuring real-time capability. For traffic surveillance scenarios, 25 FPS serves as the reference benchmark for real-time processing.

- **Number of Parameters:** The total count of model parameters, reflecting model complexity and storage requirements.
- **Temporal Stability (TS):** TS measures the frame-to-frame consistency of detection results in a video. It quantifies how smoothly detection metrics vary across consecutive frames. A TS value close to 1 indicates stable, flicker-free detection; a value close to 0 indicates severe temporal jitter or flicker caused by inconsistent enhancement across frames.

4.3 Performance of Weather Classification Module

We trained a MobileNetV3 weather classifier on the RSCM dataset, which contains 10,000 images per class for rain, fog, and snow. To evaluate the classification performance and its adaptability to traffic scenarios, we conducted tests on DAWN, a dataset specifically designed for adverse weather conditions in traffic surveillance. Table 4-3 and Table 4-4 present the evaluation metrics and confusion matrix.

Table 4-3. Performance of single-label weather classification.

Class	Precision	Recall	F1-Score	Support
Fog	80.11%	84.18%	82.10%	335
Rain	72.84%	59.00%	65.19%	200
Snow	86.55%	95.54%	90.82%	202
Weighted Avg	—	—	—	737

Table 4-4. Confusion matrix of the weather classifier.

	Predicted Fog	Predicted Rain	Predicted Snow
Actual Fog	282	42	11
Actual Rain	63	118	19
Actual Snow	7	2	193

As shown in Table 4-3 and 4-4, the single-label classifier performs well on fog (F1=82.10%) and snow (F1=90.82%), but the recall for rain is only 59.00%. Further analysis reveals that approximately 31.5% of rain samples are misclassified as fog. Inspection of these misclassified samples indicates a prevalent "rain–fog coexistence" phenomenon—water vapor during rainfall produces a haze-like global degradation in the image, causing ambiguity for the single-label classifier. This observation aligns closely with real-world weather conditions, where adverse weather often manifests in composite forms.

To address this issue, we propose a multi-label classification strategy for composite weather scenarios. During inference, a probability threshold (0.3 in this experiment) is applied: a weather class is considered present if its predicted probability exceeds the threshold. When multiple classes simultaneously exceed the threshold, the corresponding restoration networks are invoked sequentially in descending order of probability for cascaded enhancement. As shown in Table 4-5 and 4-6, under the 0.3 threshold, multi-label classification improves the rain recall from 59.00% to 75.50%, an increase of 16.5 percentage points. Furthermore, approximately 19% of test samples are identified as exhibiting composite weather, confirming the prevalence of mixed weather conditions in real-world road surveillance and validating the necessity of the multi-label design.

Table 4-5. Performance of multi-label weather classification.

Metric	Value
Overall Accuracy	80.46%
Partial Match Accuracy	88.06%
Exact Match Accuracy	70.42%
Multi-weather Rate	18.99%

Table 4-6. Comparison of single and multi-label weather classification.

Class	Single-Label Recall	Multi-Label Recall	Improvement
Fog	84.18%	89.55%	+5.37%
Rain	59.00%	75.50%	+16.50%
Snow	95.54%	98.02%	+2.48%

4.4 Performance of Individual Restoration Networks

To comprehensively evaluate the restoration capabilities of image enhancement networks under diverse adverse weather conditions, we rendered synthetic rain, fog, and snow effects onto traffic surveillance videos from the DETRAC dataset. Specifically, we segmented the source videos into standardized 5-second clips at 25 frames per second. We then randomly selected 20 clips for each weather condition, yielding a test set of 7,500 key frames to evaluate enhancement performance.

To establish performance bounds for our evaluation, we used the original clean videos as the upper bound and the synthetically degraded videos as the lower bound. This allows us to quantify the achievable performance gain from enhancement.

For rainy scenarios, we selected PReNet pre-trained on the Rain100H dataset; for foggy scenarios, we used AOD-Net pre-trained on the ITS dataset. Both networks have demonstrated excellent performance in their respective domains. For snowy weather, we adopted the TransWeather architecture and trained it from scratch for 100 epochs on the Snow100K dataset, due to the lack of a mature pre-trained model for this condition.

To provide an intuitive overview of the restoration quality, Figure 4-1 presents representative frames from all three weather conditions before and after enhancement. The visual results demonstrate that the system effectively removes weather-specific degradation patterns while preserving structural details essential for downstream analysis. Results for each weather type are reported in Sections 4.4.1 to 4.4.3.



Figure 4-1. Degraded vs. enhanced frames across rain, fog, and snow.

4.4.1 Rainy Weather Enhancement Performance

Rainy conditions substantially impaired detection. As shown in Table 4-7, high-confidence detections fell from 806.9 to 562.9 (a 30.24% decline), and average confidence dropped from 0.8483 to 0.7369. PReNet enhancement recovered high-

confidence detections to 656.0 (+16.54%) and raised mAP@0.95 from 0.0736 to 0.1680 (+128.18%), indicating that the deraining network successfully reconstructed precise target boundaries rather than merely restoring visibility.

Table 4-7. Detection performance under rain degradation and after enhancement.

Metric	Original	Degraded	After PReNet	Improvement
Avg. Confidence	0.8483	0.7369	0.7726	+4.85%
High-conf. Detections (>0.7)	806.9	562.9	656.0	+16.54%
Total Detections	885.5	770.3	810.9	+5.26%

Temporal stability also improved, with the stability score rising from 0.7932 to 0.8101. As shown in Table 4-8, rain exhibited the largest stability gain among the three weather types (+2.13%), reflecting the effectiveness of deraining in suppressing frame-to-frame fluctuations caused by random rain streak patterns.

Table 4-8. Temporal stability across weather conditions before and after enhancement.

Weather	Degraded	After Enhancement	Improvement
Rain	0.7932	0.8101	+2.13%
Fog	0.8327	0.8338	+0.13%
Snow	0.7969	0.8072	+1.29%

4.4.2 Foggy Weather Enhancement Performance

Fog caused the most severe image quality degradation among the three weather types. As shown in Table 4-9, PSNR dropped to 10.78 dB and contrast declined by 65.04%—far exceeding rain and snow. Yet high-confidence detections fell by only 25.10% (vs. 30.24% under rain), suggesting greater detector robustness to global contrast reduction than to localized rain streak interference.

AOD-Net achieved the strongest recovery (Table 4-9 and 4-10): contrast recovered by 89.66% (recovery rate 90.47%), and high-confidence detections rose to 698.6 (+15.60%). Small-target Recall improved most dramatically, from 0.4667 to 0.6123 (+31.19%), confirming that dehazing effectively restores distant vehicles obscured by atmospheric scattering.

Table 4-9. Detection performance under fog degradation and after enhancement.

Metric	Original	Degraded	After AOD-Net	Improvement
Avg. Confidence	0.8483	0.7851	0.8197	+4.41%
High-conf. Detections (>0.7)	806.9	604.4	698.6	+15.60%
Total Detections	885.5	745.7	803.0	+7.69%

Table 4-10. Image quality metrics across weather conditions after enhancement.

Metric	Rain	Fog	Snow
PSNR — Degraded (dB)	18.04	10.78	18.56
PSNR — Enhanced (dB)	22.15	13.57	25.46
Contrast Decline (Degraded)	-13.80%	-65.04%	-1.50%
Contrast Recovery (Enhanced)	+16.32%	+89.66%	-1.00%
Recovery Rate	1.10%	90.47%	5.78%

The quantitative gains in detection performance under fog are visually corroborated by Figure 4-2. The enhanced frame recovers multiple detections that the degraded frame missed, particularly for distant small vehicles. This aligns with the 31.19% improvement in small-target Recall, confirming that dehazing restores semantically meaningful information critical for downstream recognition.

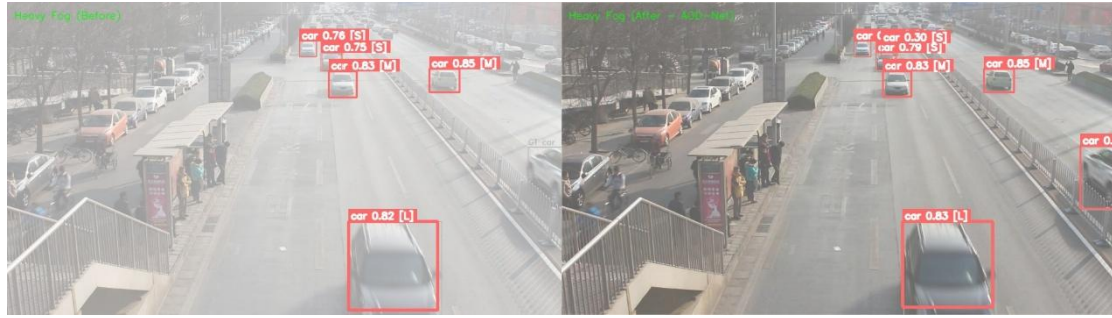


Figure 4-2. YOLO detection under fog: degraded (left) vs. enhanced (right).

A particularly instructive finding concerns the relationship between pixel-level and semantic-level metrics. As shown in Table 4-11, fog enhancement produced a strongly negative noise suppression rate of -234.86%, yet all downstream detection metrics registered consistent gains. By contrast, rain and snow exhibited positive noise suppression with negligible recovery rates. This divergence reveals the pixel-semantic discrepancy: a systematic misalignment in which pixel-level fidelity metrics do not correlate positively with task-level performance. The high-frequency components introduced during dehazing, although statistically classified as noise, in fact encode semantically meaningful edge and texture information that directly benefits detection.

Table 4-11. Noise suppression rate and recovery rate across weather conditions.

Metric	Rain	Fog	Snow
Noise Suppression Rate	43.37%	-234.86%	69.56%
Recovery Rate	1.10%	90.47%	5.78%

4.4.3 Snowy Weather Enhancement Performance

Snow exhibited fundamentally different degradation characteristics. Contrast declined by only 1.50% (comparing to 13.80% for rain and 65.04% for fog), confirming that snow degradation is dominated by localized occlusion rather than global attenuation. High-confidence detections fell by 26.09%, placing snow between rain and fog in impact severity.

TransWeather enhancement yielded measurable but modest recovery: high-confidence detections rose to 650.1 (+9.01%), and mAP@0.5 improved to 0.8209 (+7.02%)—substantially smaller gains than under rain and fog.

Table 4-12. Detection performance under snow degradation and after enhancement.

Metric	Original	Degraded	After TransWeather	Improvement
Avg. Confidence	0.8483	0.7723	0.7932	+2.71%
High-conf. Detections	806.9	596.4	650.1	+9.01%
Total Detections	885.5	759.4	782.8	+3.07%

The most notable characteristic of snow enhancement emerges from the multi-scale Recall breakdown. As Table 4-13 shows, large-target Recall improved markedly (+10.32%), whereas small-target Recall improved only marginally (+3.16%). Figure 4-3 provides visual confirmation of this asymmetry: detection confidence recovers visibly for larger vehicles, while small and heavily occluded targets remain largely unaffected. We attribute this to two factors. First, when large snowflakes completely cover key regions of small targets, the occluded information is permanently lost during imaging and cannot be recovered from a single frame. Second, the desnowing process may inadvertently suppress fine-grained features of small objects, as these pixel-level details can resemble snow particles and be removed alongside actual snow. Large targets, by contrast, retain sufficient visible regions even under partial occlusion and are less vulnerable to feature loss during restoration. This dual limitation delineates a fundamental boundary for single-frame snow enhancement and suggests temporal fusion as a direction for future work.

Table 4-13. Multi-scale Recall under snow degradation and after enhancement.

Target Size	Degraded	After TransWeather	Improvement
Small	0.6544	0.6751	+3.16%
Medium	0.8978	0.9161	+2.03%
Large	0.7646	0.8435	+10.32%



Figure 4-3. YOLO detection under snow: degraded (left) vs. enhanced (right).

4.5 Ablation Study

This section presents two ablation experiments designed to validate the core design decisions of the RoadClear system: (1) the necessity of weather classification with dedicated restoration networks, and (2) the overall benefit of enhancement for downstream detection.

(1) Ablation 1: Necessity of Weather-specific Restoration Networks.

To validate the necessity of the weather classification and scheduling module, we compare the full pipeline against three single-network baselines in which a single restoration network—PReNet, AOD-Net, or TransWeather—is applied regardless of the actual weather condition. The experimental setup, including the test video sequences and evaluation protocol, is identical to that described in Section 4.4. Table 4-14 reports $\text{mAP}@0.5$ for all configurations across rain, fog, and snow.

Table 4-14. Ablation study on weather-specific restoration networks ($\text{mAP}@0.5$).

Weather	PReNet	AOD-Net	TransWeather	Full Pipeline (Ours)
Rain	0.815	0.426	0.618	0.815
Fog	0.747	0.881	0.812	0.881
Snow	0.704	0.473	0.821	0.821

The results reveal two key findings. First, dedicated networks achieve optimal performance on their corresponding weather types. The full pipeline matches the best single-network baseline in every case, confirming correct dispatch by the classification module.

Second, cross-domain application causes severe degradation. The most extreme case occurs when AOD-Net is applied to snowy scenes, where $\text{mAP}@0.95$ falls from 0.249 to 0.062—a 75.0% decline. Applying AOD-Net to rain reduces $\text{mAP}@0.5$ by 47.7%, while applying PReNet to fog lowers $\text{mAP}@0.95$ by 59.6%. Notably, the dehazing network proves most disruptive off-domain, as its global contrast

manipulation severely distorts non-hazy images, whereas the deraining network's localized filtering causes comparatively milder degradation.

These findings confirm that weather degradation patterns are fundamentally distinct and a single network cannot generalize across weather types, validating the core design rationale of RoadClear: weather classification with type-specific enhancement.

(2) Ablation 2: Impact of Enhancement on Downstream Detection.

To quantify the practical benefit of the enhancement pipeline, we compare detection performance with and without enhancement on degraded videos. Table 4-15 summarizes the results across all three weather conditions using two representative metrics: mAP@0.5 and the number of high-confidence detections (confidence > 0.7).

Table 4-15. Impact of enhancement on downstream detection performance.

Weather	Metric	Without Enhancement	With Enhancement	Relative Improvement
Rain	mAP@0.5	0.701	0.815	+16.2%
	High-conf. detections	562.9	656.0	+16.5%
Fog	mAP@0.5	0.795	0.881	+10.8%
	High-conf. detections	604.4	698.6	+15.6%
Snow	mAP@0.5	0.767	0.821	+7.0%
	High-conf. detections	596.4	649.7	+8.9%

Enhancement yields consistent gains across all weather types. The most pronounced improvements are observed under rain, where mAP@0.5 increases by 16.2% and high-confidence detections rise by 16.5%. The detailed per-weather analysis, including mAP@0.95, stability metrics, and the pixel-semantic discrepancy phenomenon, is presented in Section 4.4.2.

Taken together, the two ablation studies confirm that (1) the classification-driven multi-network architecture is superior to any single-network alternative, and (2) the enhancement module provides measurable and consistent downstream benefits across all tested weather conditions.

4.6 Chapter Summary

This chapter presented RoadClear, a video enhancement and scheduling system integrating weather classification, dedicated restoration networks, and downstream detection evaluation. Extensive experiments on synthetically degraded videos demonstrated the effectiveness of the proposed approach. Per-weather analysis showed consistent detection improvements across all conditions, most notably under fog (small-target Recall +31.2%) and rain (mAP@0.95 +128.2%). The pixel-semantic discrepancy observed under fog revealed the misalignment between pixel-level and task-level metrics. Ablation studies confirmed the necessity of weather-specific networks—cross-domain application caused up to 75.0% mAP@0.95 degradation—and the overall benefit of enhancement.

These results establish RoadClear as an effective solution for mixed-weather traffic surveillance, while limitations in snow occlusion and license plate recognition point to directions for future work.

CHAPTER 5: SYSTEM DESIGN AND DEVELOPMENT

5.1 Requirement Analysis

Before system design, a comprehensive analysis of the functional and non-functional requirements of the RoadClear system is needed to clarify the capabilities and performance indicators the system should possess.

5.1.1 Functional requirements

1) The user can upload videos in multiple formats. The system shall assign a unique ID to each video and support video query, preview, download, deletion, and status viewing.

2) The system shall automatically recognize weather types and output probability distributions; users may manually override results, and all recognition logs shall be recorded for analysis.

3) The system shall automatically select the appropriate enhancement model, support single or cascaded enhancement modes, provide real-time progress feedback, and allow batch processing.

4) The system shall evaluate enhancement effect via object detection without manual annotation, generate multi-metric reports with visual comparisons, and support side-by-side video playback.

5) Users can adjust inference parameters (e.g., frame-skipping, post-processing); preset schemes shall be provided, with real-time parameter and persistent storage.

5.1.2 Non-functional requirements

1) The system shall process video at ≥ 10 FPS (640×480), classify weather within 150ms per frame, start up within 10 seconds (including model loading), keep memory usage ≤ 4 GB per video.

2) The system shall have high availability, with graceful degradation on abnormal frames without interrupting the overall processing flow.

3) The system shall provide an intuitive Web interface with operation response ≤ 200 ms, real-time progress and remaining time display, and clear error messages with actionable suggestions.

4) The system shall adopt a modular design with low coupling, support plug-and-play addition of new enhancement models, expose standardized RESTful APIs for third-party integration, and allow configuration adjustments without code changes.

5.2 System Context

Based on requirement analysis, the RoadClear system adopts a front-end and back-end separation architecture design, with the back-end responsible for video processing and model inference, and the front-end providing user interaction interface.

5.2.1 Overall Architecture

RoadClear adopts a modular architecture comprising three main components: the frontend, the backend service, and the model pool. Figure 5-1 illustrates the overall system architecture and the communication mechanisms between components.

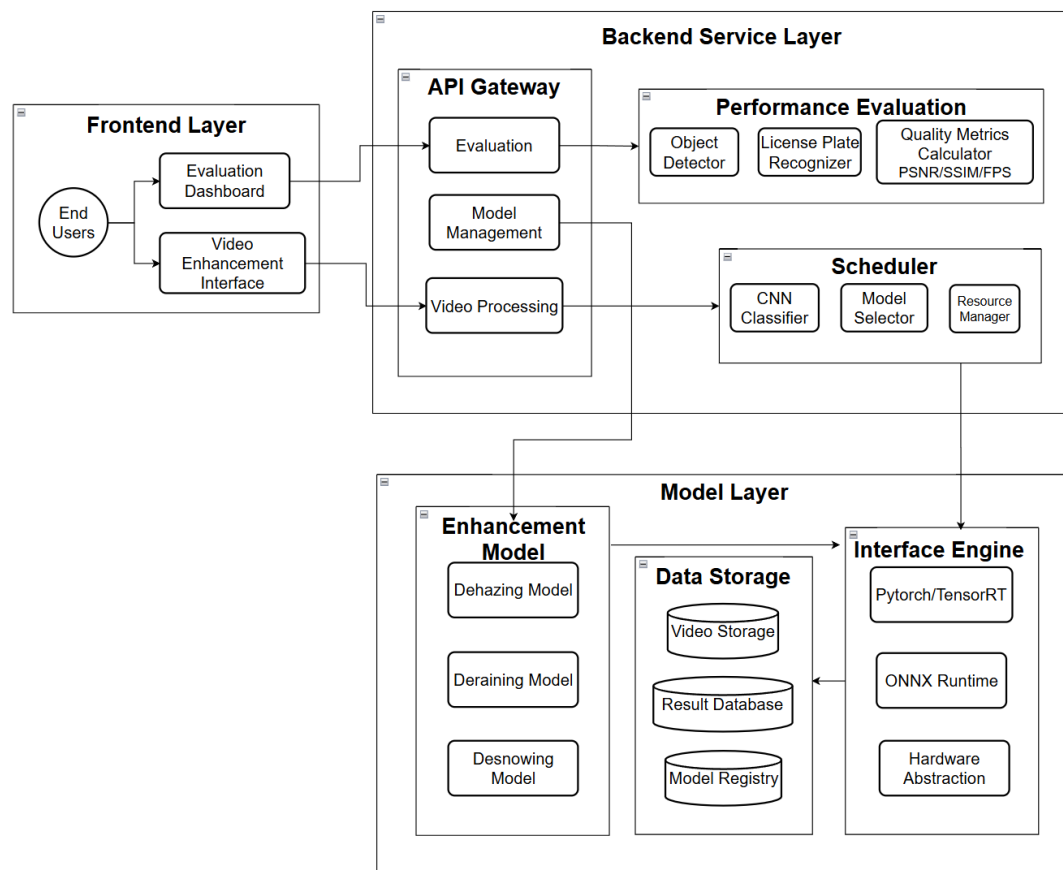


Figure 5-1. System Architecture of RoadClear System

The frontend, built with Vue.js 3 and Element Plus, provides a browser-based interface for video upload, model selection, result comparison, and evaluation report

visualization. It communicates with the backend exclusively through RESTful APIs over HTTP.

The backend service, implemented with FastAPI, hosts the core processing pipeline. Five modules collaborate to execute the enhancement workflow: the Scheduler Module orchestrates the overall process by invoking the Video Module for frame acquisition, the Model Module for restoration network access, the Evaluation Module for downstream detection, and the Config Module for runtime parameters. All deep learning inference is accelerated via ONNX Runtime.

The model pool manages the restoration networks (AOD-Net, PReNet, TransWeather) and the downstream YOLOv8 detectors. Models are dynamically loaded and unloaded by the backend service as needed, and are stored alongside ONNX exports and configuration files on the local file system..

5.2.2 Module Interaction Flow

The system modules interact through HTTP and internal function calls. The primary workflow is illustrated in Figure 5-2. The user first uploads a video through the frontend; the backend Video Module saves it, assigns a unique video ID, and returns the ID to the frontend. Upon receiving the enhancement request with model selection, the backend initiates per-frame processing: the Scheduler Module performs weather classification every 30 frames, retrieves the appropriate restoration model from the Model Pool, runs enhancement, optionally applies post-processing, and writes the enhanced frame to the output video. The frontend polls the status endpoint for real-time progress updates and displays the comparison upon completion.

Two additional workflows complement the main pipeline. The evaluation workflow is triggered when the user requests an assessment: the Evaluation Module runs the YOLOv8 detector on both original and enhanced videos, computes image quality metrics (such as PSNR, SSIM) and temporal stability, and generates an HTML evaluation report with side-by-side visual comparisons and quantitative results. This enables systematic validation of enhancement performance.

The configuration workflow handles runtime parameter adjustments: the user modifies settings such as the frame-skipping interval via the frontend, and the Config Module validates, applies, and persists the new parameters to a JSON file.

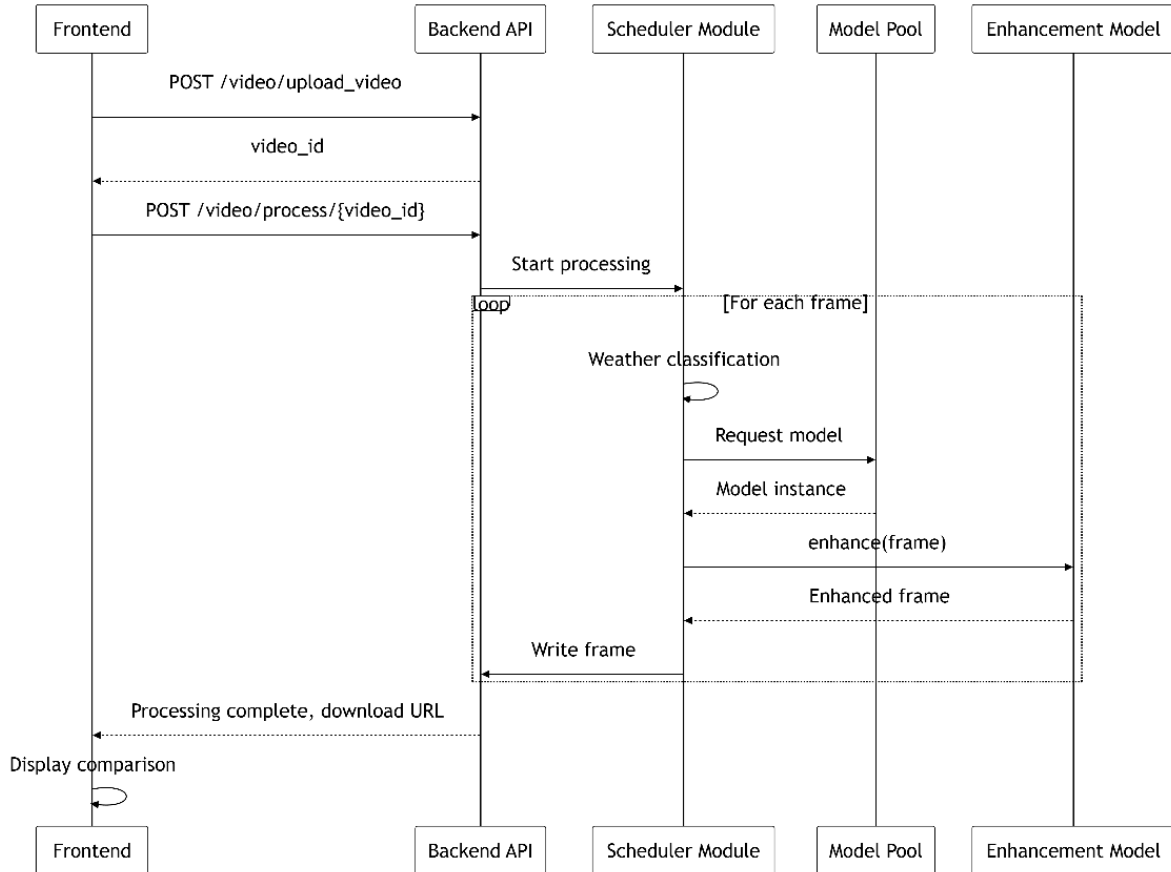


Figure 5-2. Sequence Diagram of RoadClear System

5.3 System Interface

The RoadClear system provides an intuitive and user-friendly web interface, allowing users to access various system functions through a browser.

5.3.1 Main Interface

The RoadClear frontend is a single-page application built with Vue 3 (Composition API) and Element Plus, bundled by Vite, with Axios handling HTTP communication. The main interface adopts a tab-based layout comprising five modules: IntroPage, VideoEnhancement, AlgorithmShowcase, ParameterConsole, and VideoEvaluation. Each module is encapsulated as an independent Vue component.

The IntroPage (Figure 5-3) displays a carousel showcasing the enhancement results of each algorithm, providing users with an immediate visual impression of the system's restoration capabilities before they begin processing their own videos.

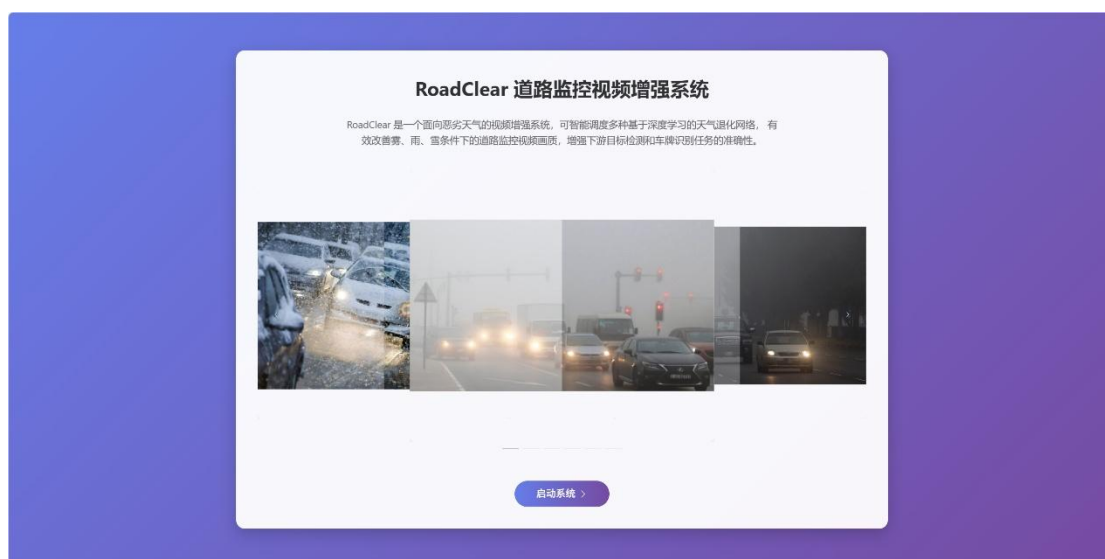


Figure 5-3 IntroPage of RoadClear System

5.3.2 Video Enhancement Interface

The VideoEnhancement tab, shown in Figure 5-4, is the primary interface for video upload and processing. Users upload traffic surveillance footage and select a restoration model from the following options: Auto (weather-adaptive, the default), AOD-Net (dehazing), PreNet (deraining), TransWeather (desnowing), Simple (lightweight bilateral filtering), or a user-defined cascaded combination.



Figure 5-4 Video Enhancement Tab of RoadClear System(a)

As shown in Figure 5-5, A real-time progress bar and log panel provide processing feedback, alongside a side-by-side display of the original and enhanced videos. The enhanced video can be downloaded upon completion.

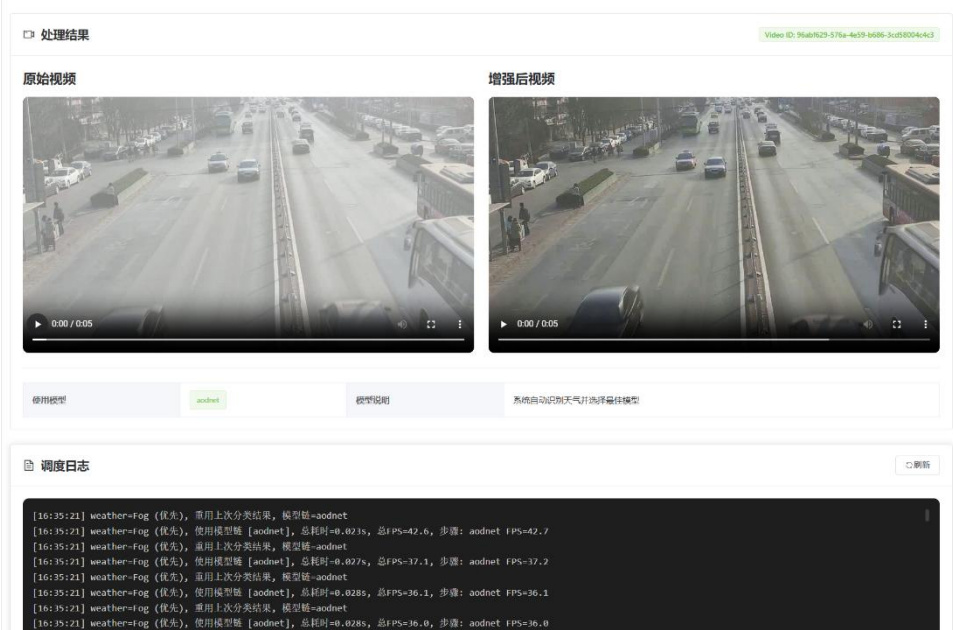


Figure 5-5 Video Enhancement Tab of RoadClear System(b)

5.3.3 Algorithm Showcase Interface

The AlgorithmShowcase tab provides visual explanations of each enhancement algorithm. As illustrated in Figure 5-6 to 5-8, it displays feature maps, residual maps, and GradCAM overlays alongside side-by-side before-and-after comparisons, helping users understand the working principles of the restoration networks.

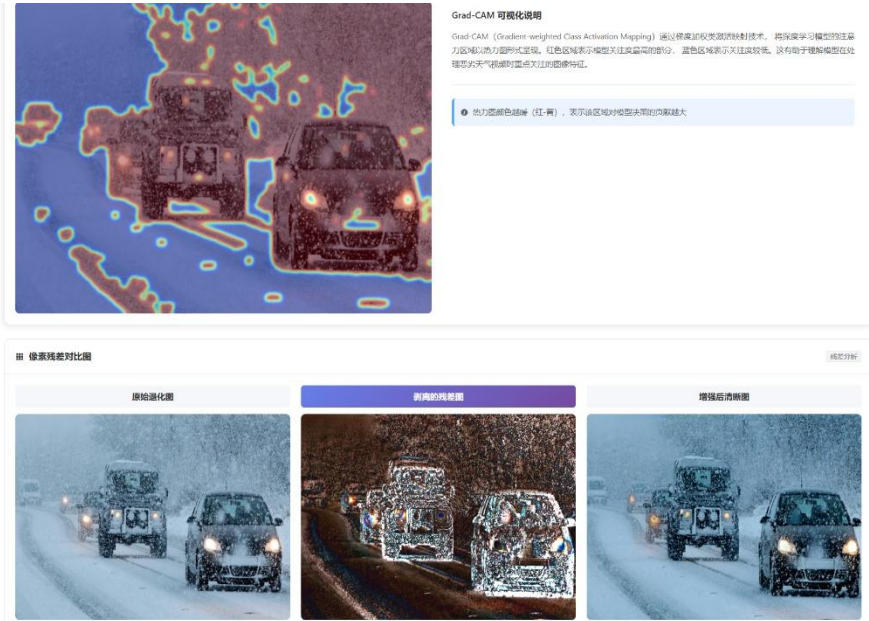


Figure 5-6 AlgorithmShowcase Tab of RoadClear System(a)

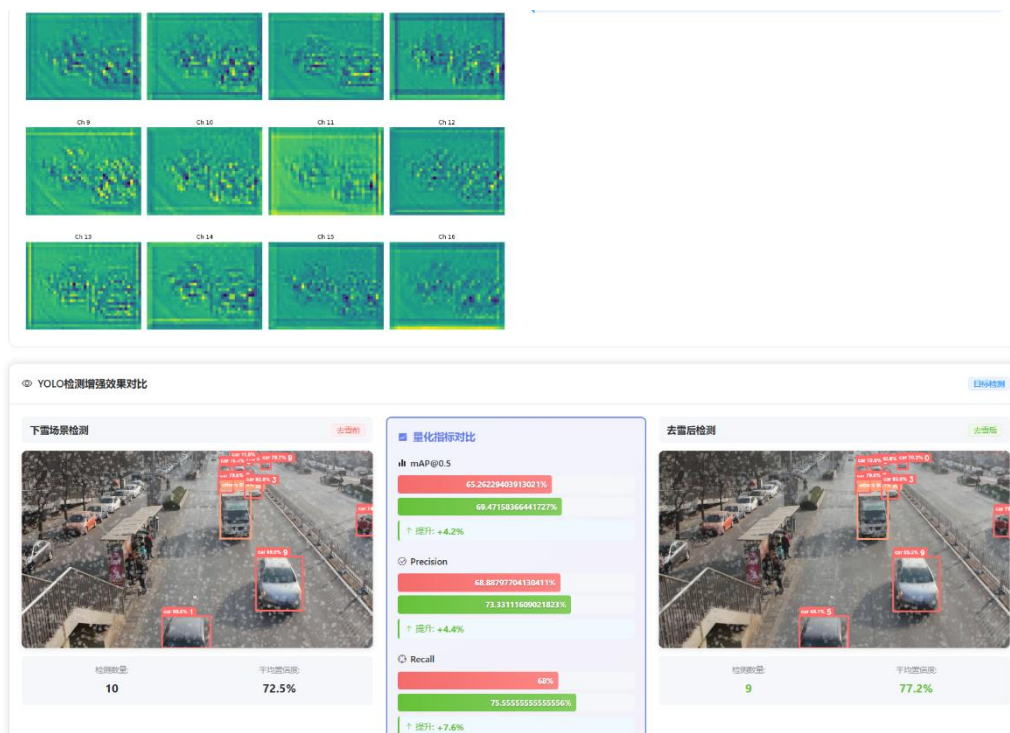


Figure 5-7 AlgorithmShowcase Tab of RoadClear System(b)



Figure 5-8 AlgorithmShowcase Tab of RoadClear System(c)

5.3.4 Parameter Console Interface

The ParameterConsole tab enables runtime adjustment of system settings such as the frame-skipping interval and post-processing toggles. Changes are validated and pushed to the backend in real time. Figures 5-9 to 5-11 illustrate the Parameter Console Interface.

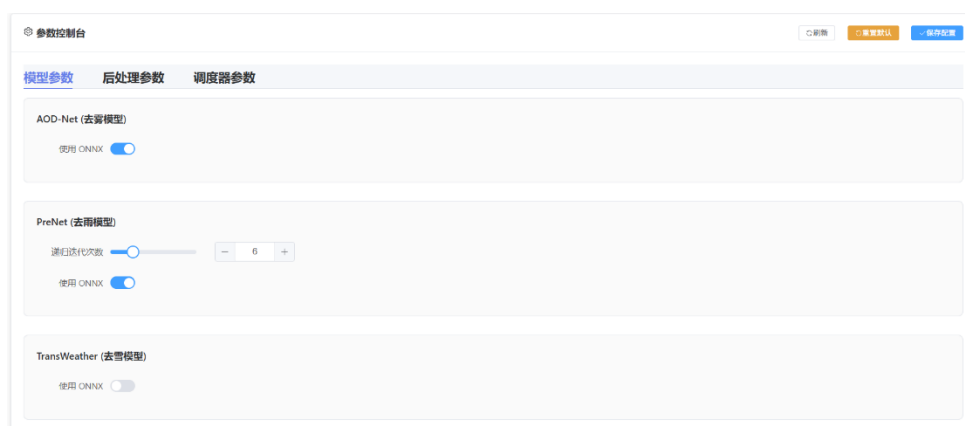


Figure 5-9 ParameterConsole Tab of RoadClear System(a)

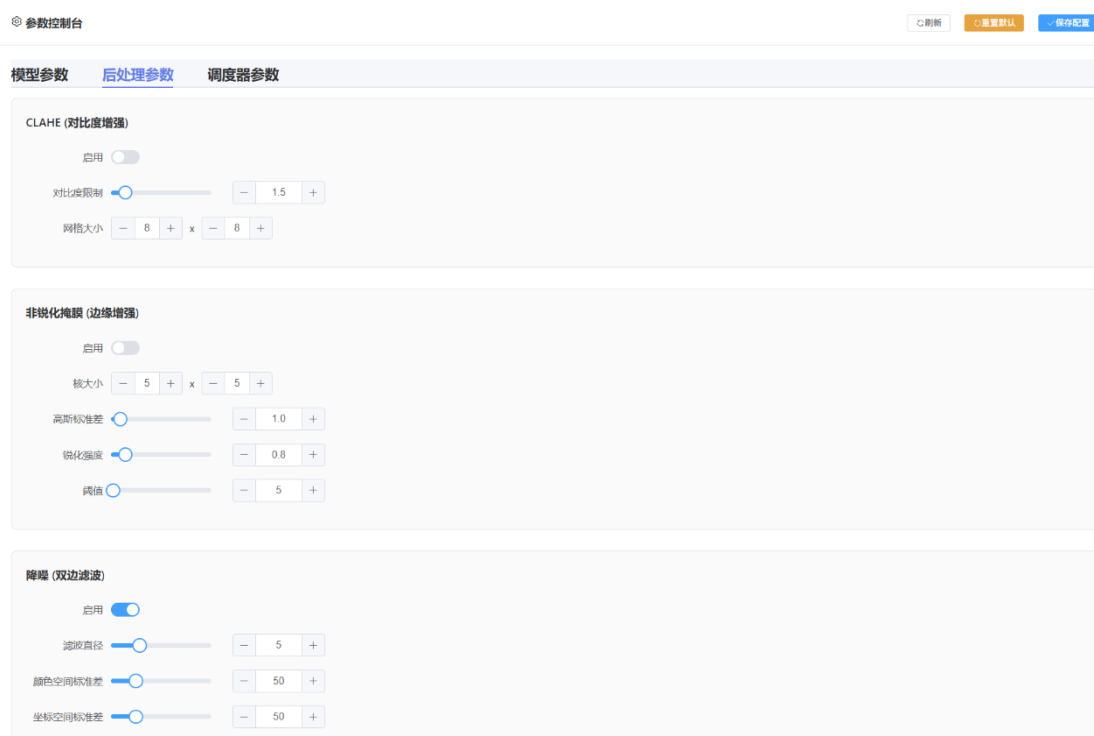


Figure 5-10 ParameterConsole Tab of RoadClear System(b)



Figure 5-11 ParameterConsole Tab of RoadClear System(c)

5.3.5 Evaluation Interface

The VideoEvaluation tab, shown in Figure 5-12 and 5-13, integrates YOLOv8 detectors to compare detection performance before and after enhancement. Results are displayed through YoloComparisonCard components with confidence distributions and bounding box counts, and a downloadable HTML evaluation report can be generated.

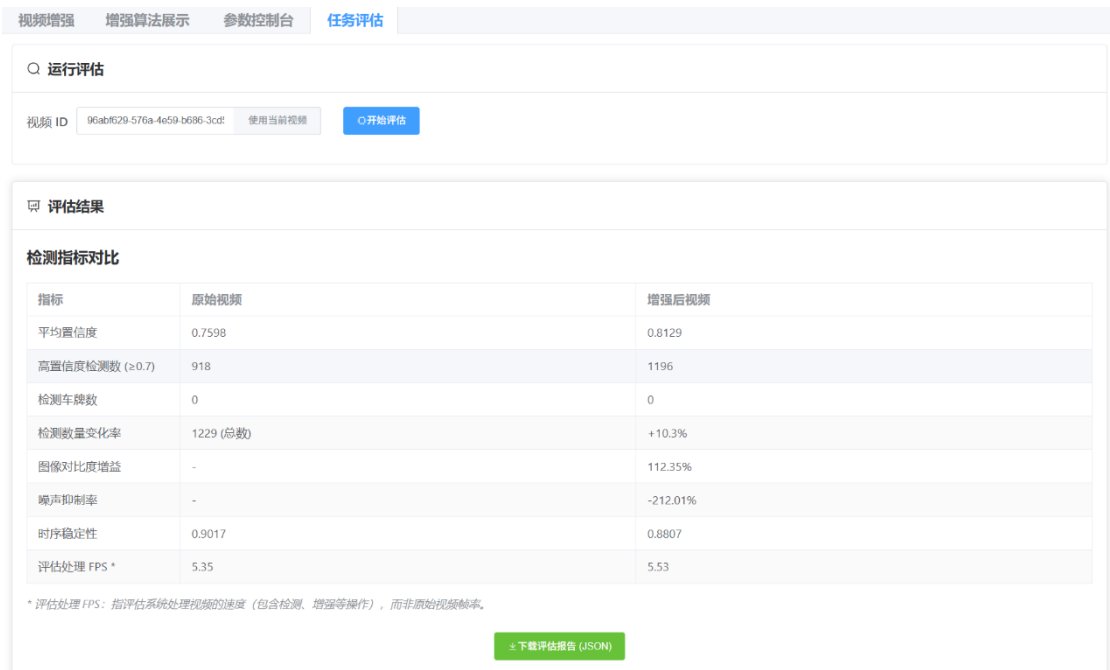


Figure 5-12 VideoEvaluation Tab of RoadClear System(a)



Figure 5-13 VideoEvaluation Tab of RoadClear System(b)

5.4 Chapter Summary

This chapter presented the design and implementation of the RoadClear system. A requirement analysis was first conducted to clarify the system's functional and non-functional requirements. The system adopts a front-end and back-end separation three-tier architecture, with Vue.js 3 and Element Plus for the presentation layer, FastAPI for the RESTful API service layer integrating PyTorch, TensorFlow, and ONNX Runtime, and a file-system-based data layer.

This chapter describes the interaction flows among modules, covering key processes including video upload and enhancement, effect evaluation, and parameter configuration. Finally, the main system interfaces were demonstrated, including the video enhancement page, effect evaluation page, algorithm showcase page, and evaluation report visualization components.

CHAPTER 6: CONCLUSION AND FEATURE WORK

6.1 Conclusion

This paper presented RoadClear, an intelligent video enhancement system for traffic surveillance under adverse weather conditions. The system integrates a lightweight MobileNetV3-Small weather classifier, an inter-frame skipping scheduler that dynamically routes frames to weather-specific restoration networks (AOD-Net for fog, PReNet for rain, TransWeather for snow), and a downstream YOLOv8-based detection evaluation pipeline. All inference is accelerated via ONNX Runtime, with temporal consistency processing ensuring stable output. The system is deployed as a B/S-architecture visualization platform built with Vue and FastAPI.

The weather classifier was trained on the RSCM dataset and evaluated on the DAWN benchmark. Under single-label classification, it achieved F1-scores of 82.10% for fog and 90.82% for snow, while rain recall reached only 59.00%. Further analysis revealed a prevalent rain–fog coexistence phenomenon in which water vapor during rainfall produces haze-like global degradation, causing classifier ambiguity. To address this, a multi-label strategy with a 0.3 probability threshold was introduced, raising rain recall to 75.50%, a gain of 16.5 percentage points. Approximately 19% of test samples were identified as exhibiting composite weather, confirming the practical necessity of multi-label classification for real-world surveillance.

Extensive experiments on synthetically degraded DETRAC videos demonstrated consistent downstream detection improvements. Under fog, small-target Recall increased from 0.4667 to 0.6123, a gain of 31.19%. Under rain, mAP@0.95 surged from 0.0736 to 0.1680, an increase of 128.18%. Under snow, high-confidence detections rose from 596.4 to 650.1, a gain of 9.01%. Ablation studies confirmed the necessity of weather-specific networks: cross-domain application of a single restoration model caused up to 75.0% degradation in mAP@0.95. The overall benefit of enhancement for detection was also verified.

Two notable findings emerged from the experiments. First, under foggy conditions, we observed a pixel-semantic discrepancy: while the enhanced images appeared noisier at the pixel level, all downstream detection metrics improved consistently. This suggests that traditional pixel-level quality metrics do not always reflect the practical value of enhancement for recognition tasks. Second, under snowy conditions, large

targets benefited noticeably from enhancement while small targets showed only marginal gains. This asymmetry points to a fundamental limitation of single-frame methods: when snowflakes fully cover key regions of small objects, the lost information cannot be recovered from a single image.

6.2 Feature Work

Several directions remain open for future investigation.

(1) Temporal fusion for severe occlusion.

The physical irreversibility of snowflake occlusion identified in this work suggests that single-frame enhancement has inherent limits. Incorporating temporal information fusion across consecutive frames or multi-frame super-resolution techniques may help recover details lost to severe occlusion.

(2) Practical deployment of license plate recognition.

The license plate recognition module currently serves only as a proof-of-concept due to the low resolution and overhead camera angle in the test environment. Integrating camera-level digital zoom or dedicated high-resolution region-of-interest cropping could substantially improve its practical utility.

(3) Evaluation of post-processing modules.

The optional post-processing module, including CLAHE, unsharp masking, bilateral filtering, and dynamic range compression, has been designed but not experimentally evaluated. Its impact on downstream task performance remains to be quantified.

(4) Extension to nighttime and mixed weather.

Extending the system to handle nighttime and low-light adverse weather conditions, as well as expanding the weather taxonomy to include mixed-type conditions such as fog-rain or snow-fog composites, represents a natural progression of this work.

Reference

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